

Summary Data on Acid Cropland and Lime Requirements in Kenya

GAIA

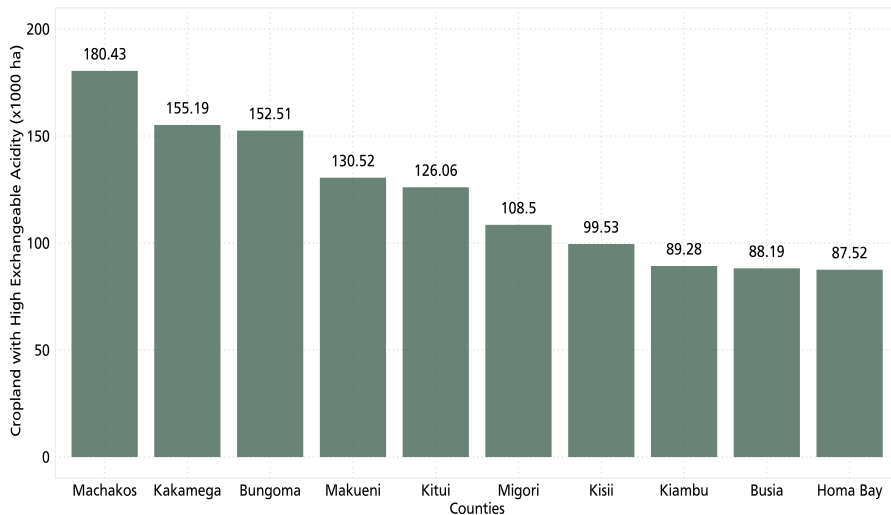
www.acidsoils.africa

Last Updated: August 8, 2024

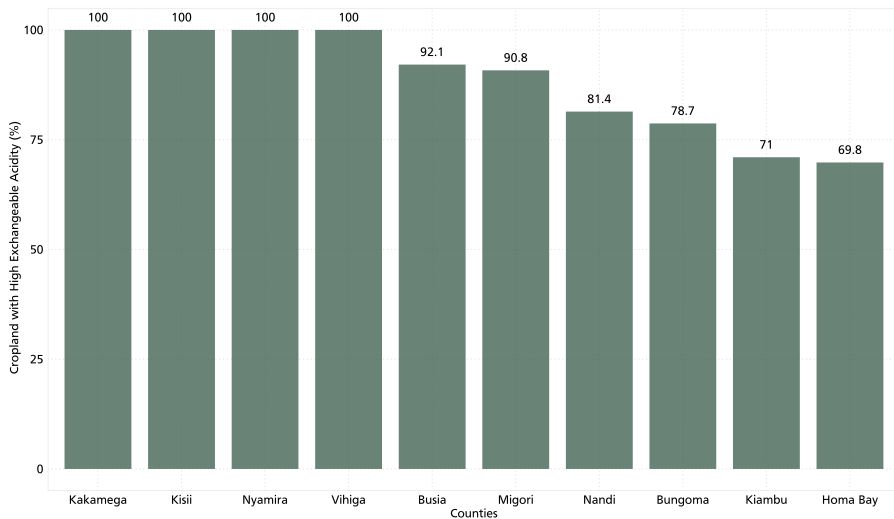
Outlook

- 1 Introduction
- 2 Acidic Cropland
- 3 Yield Loss
- 4 Lime Requirement
- 5 Yield Response
- 6 Additional Production
- 7 ROI for Lime Application
- 8 Profitability of Lime Application

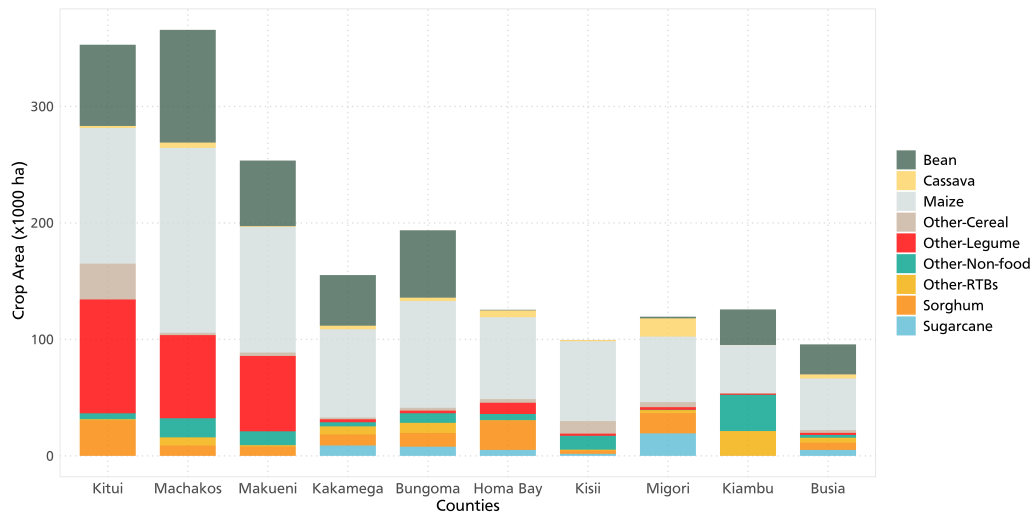
Top 10 counties in Kenya with the highest acidic cropland



Top 10 counties in Kenya with the highest acidic cropland (Percentage of total cropland)



Total crop area in the top-10 most acidic counties, not segregated by acidic cropland



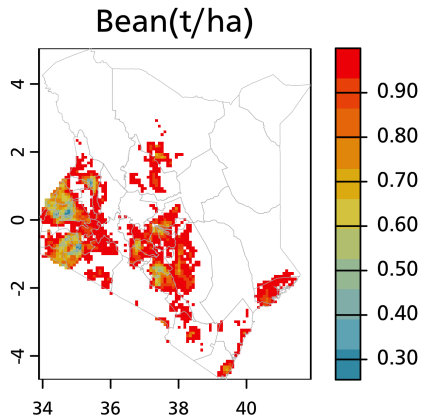
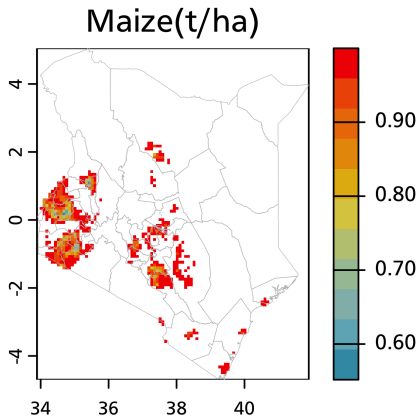
KENYA-Summary

└ Total crop area distribution (ha)

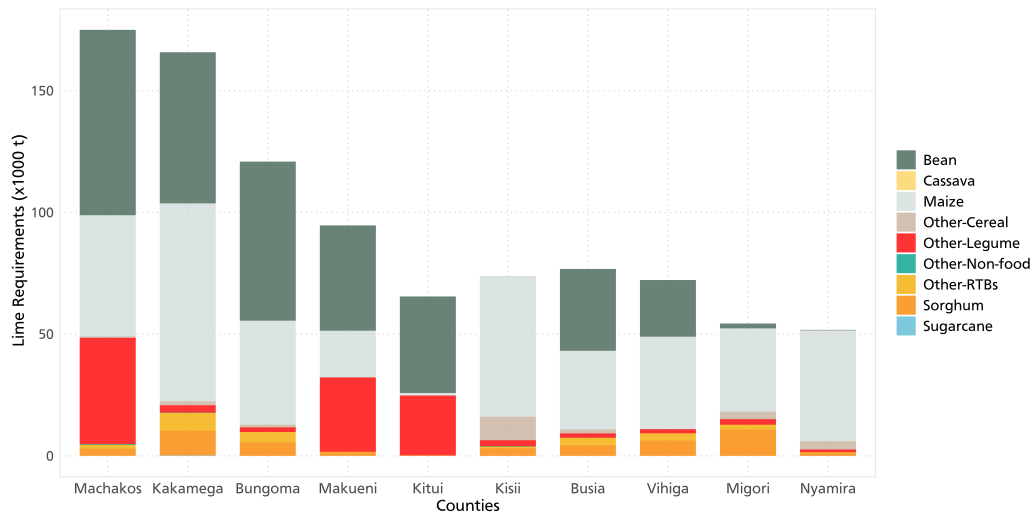
└ Total crop area in the top-10 most acidic counties, not segregated by acidic cropland

County	Maize	Sorghum	Other-Cereal	Bean	Other-Legume	Cassava	Sugarcane	Other-RTBs	Other-Non-food
Bungoma	91.63	11.74	2.46	57.88	2.17	2.91	7.87	8.79	8.25
Busia	44.29	6.16	2.39	25.79	1.76	3.50	5.13	4.28	2.42
Homa Bay	70.27	25.39	3.15	0.46	9.63	5.94	5.10	0.17	5.32
Kakamega	75.78	9.53	1.50	43.44	2.48	3.07	8.91	6.89	3.59
Kiambu	41.28	0.09	0.47	30.46	0.92	0.13	-	21.24	31.11
Kisii	68.52	2.99	10.79	0.09	1.86	1.07	1.70	0.63	11.88
Kitui	116.53	30.68	30.68	69.77	97.76	1.70	-	0.78	5.08
Machakos	158.79	8.94	1.99	96.77	71.39	4.52	-	6.98	16.36
Makueni	107.91	8.16	2.99	56.32	64.68	0.55	-	1.08	11.84
Migori	56.27	17.37	4.37	1.41	2.25	15.64	19.29	2.60	0.29

Yield loss due to soil acidity for Maize and Beans in Kenya



Lime requirements (t) for acid soil remediation in Kenya

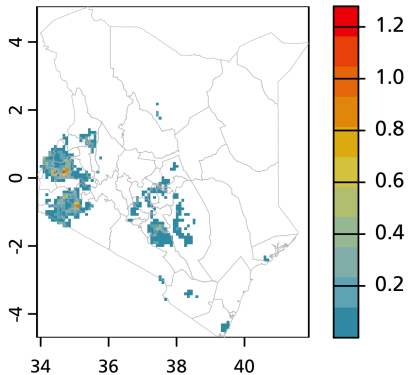


Lime requirements (t) for acid soil remediation in Kenya (table)

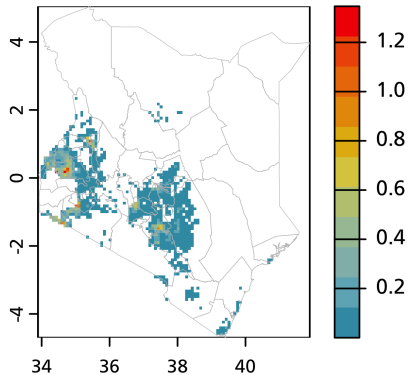
County	Maize	Sorghum	Other-Cereal	Bean	Other-Legume	Cassava	Sugarcane	Other-RTBs	Other-Non-food
Bungoma	42.71	5.55	1.12	65.36	1.90	-	-	4.27	-
Busia	32.27	4.35	1.67	33.60	1.80	-	-	3.07	0.03
Kakamega	81.26	10.17	1.62	62.05	3.01	0.06	0.18	7.30	0.14
Kisii	57.36	3.16	9.66	0.12	2.49	-	-	0.62	0.18
Kitui	1.00	0.20	0.07	39.62	24.58	-	-	0.00	-
Machakos	49.81	2.97	0.54	76.14	43.67	-	-	1.56	0.31
Makueni	19.01	1.41	0.21	43.27	30.51	-	-	0.26	-
Migori	34.20	10.69	3.12	2.01	2.25	-	-	2.12	-
Nyamira	45.56	1.09	3.37	0.19	1.15	-	-	0.39	0.01
Vihiga	37.92	6.28	0.11	23.27	1.67	-	-	3.01	-

Yield response to lime application in Kenya

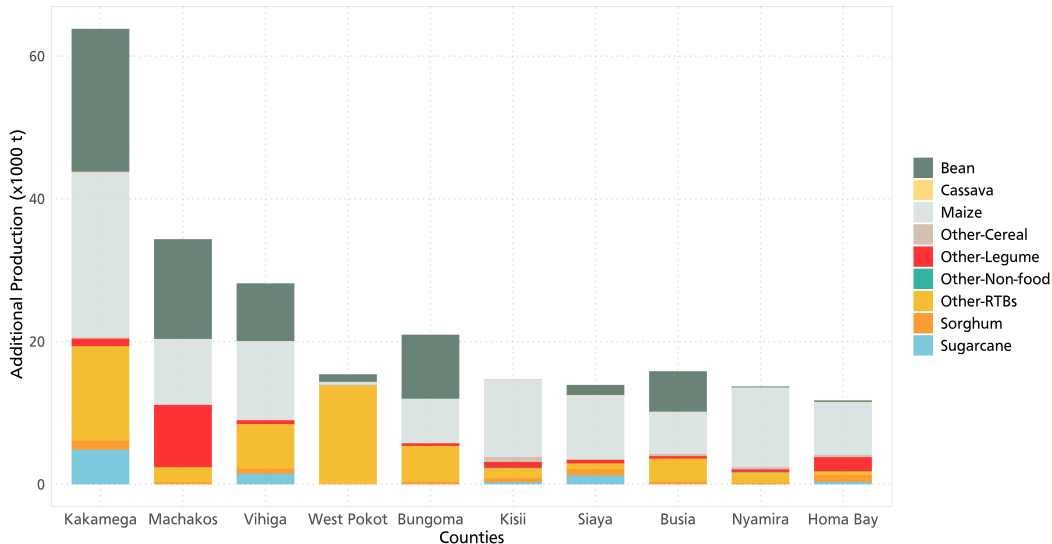
Maize(t/ha)



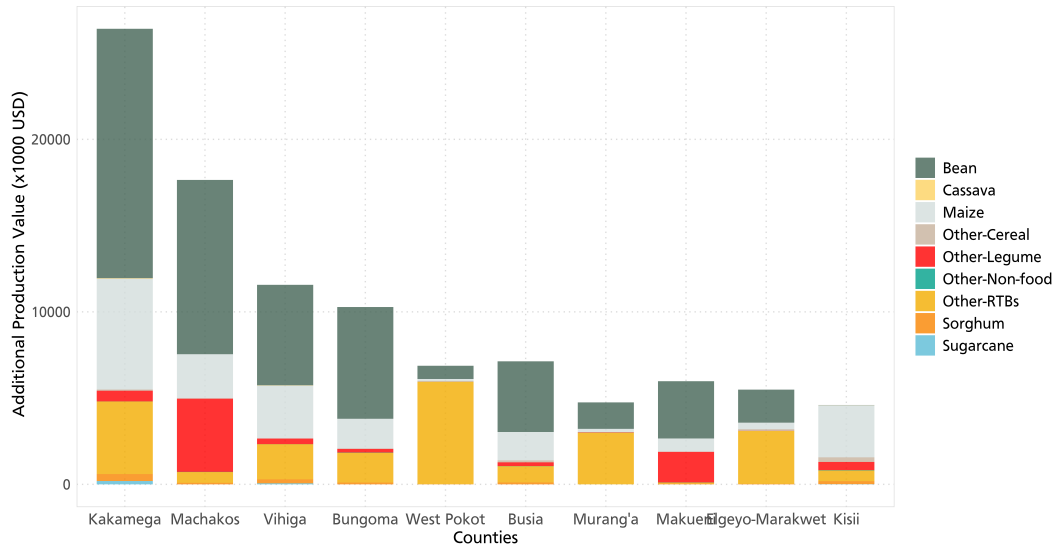
Bean(t/ha)



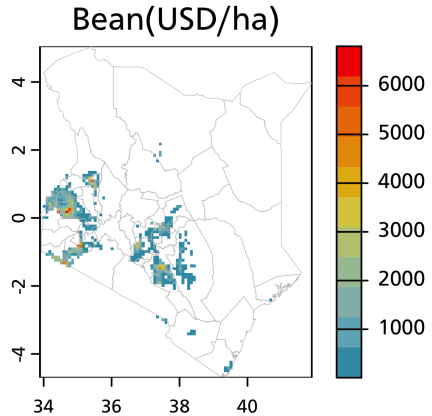
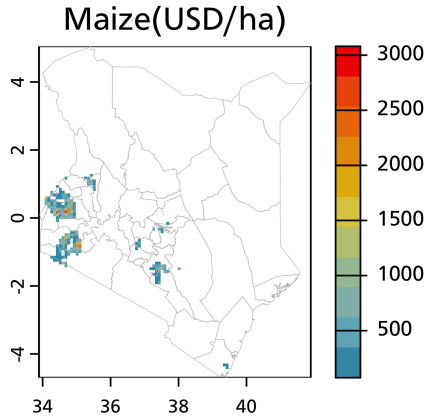
Potential for additional production (tone) in top 10 acidic counties in Kenya



Potential for additional production value (US\$) in top 10 acidic counties in Kenya



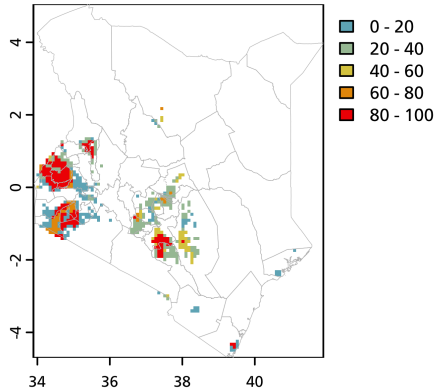
Return on investment for lime application in Kenya



- └ Profitability of Lime Application
- └ Profitability of lime application in Kenya

Profitability of lime application in Kenya

Profitable area (%)



Weighted profit (USD/ha)

